

*64. Four positive and four negative point charges of equal magnitudes $\pm Q$ are arranged alternately on the corners of a cube of edge d (see Fig. 25.40). What is the electric energy of this arrangement?

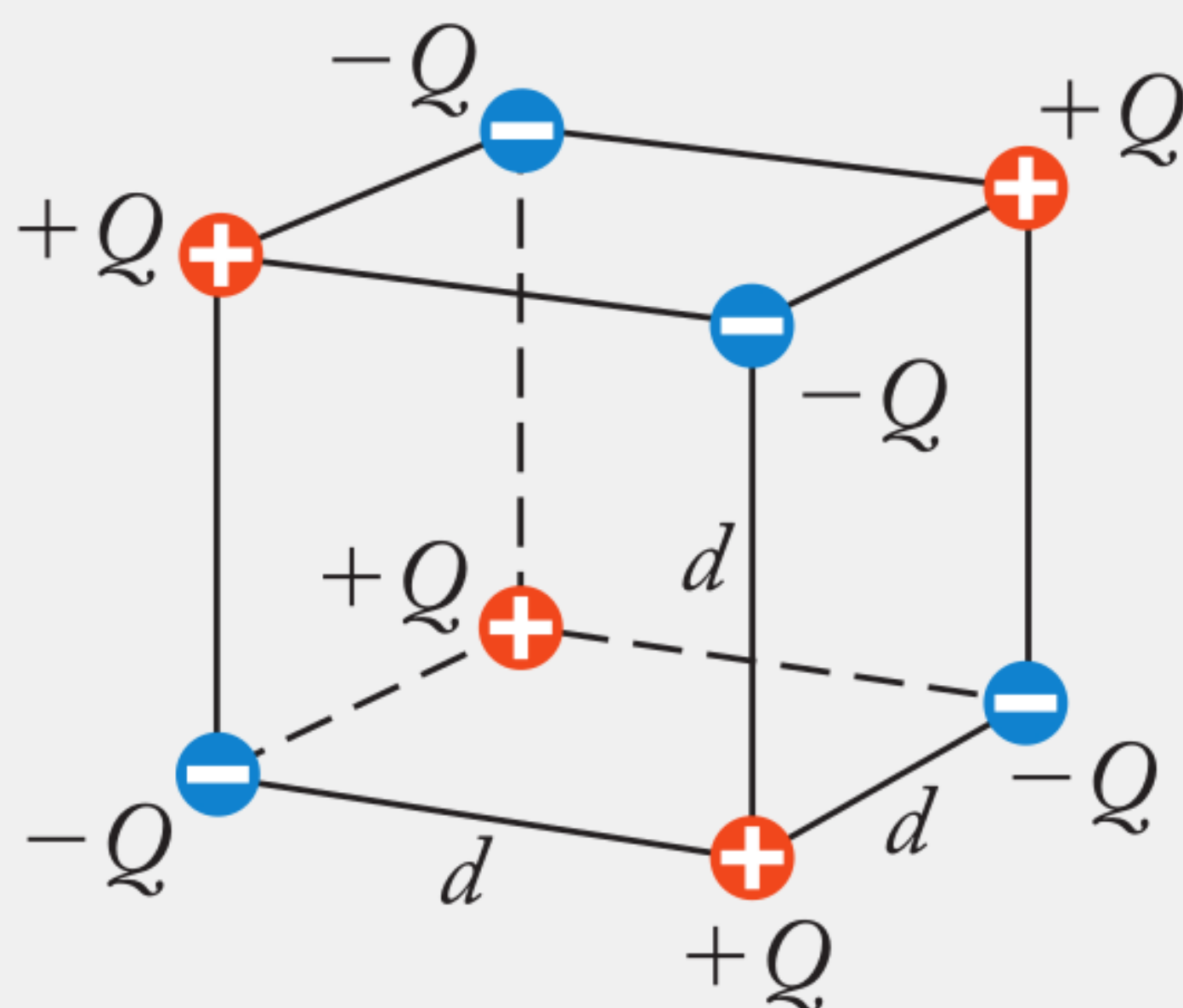


FIGURE 25.40 Point charges at the corners of a cube.

*89. Two thin rods of length l carry equal charges Q uniformly distributed over their lengths. The rods are aligned, and their nearest ends are separated by a distance x (Fig. 25.46). Calculate the mutual electric potential energy. Ignore the self-energy of each rod.

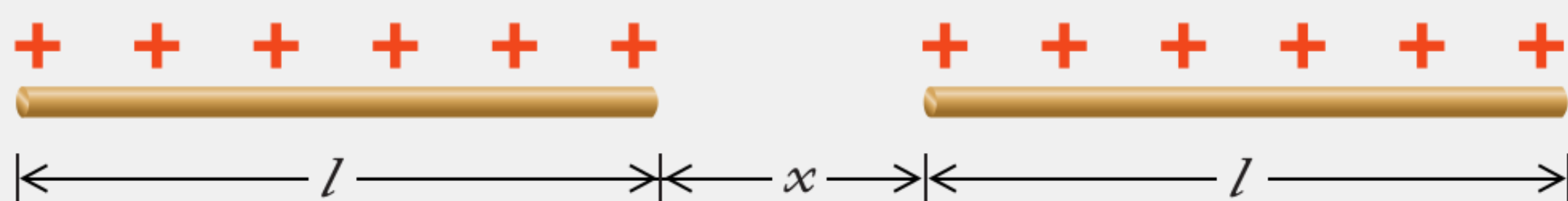


FIGURE 25.46 Two aligned charged rods.

- *26. A multiplate capacitor, such as used in radios, consists of four parallel plates arranged one above the other as shown in Fig. 26.30. The area of each plate is A , and the distance between adjacent plates is d . What is the capacitance of this arrangement?

